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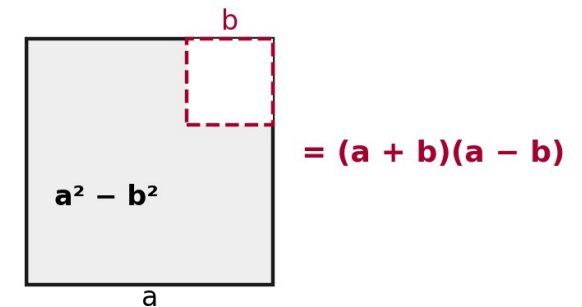
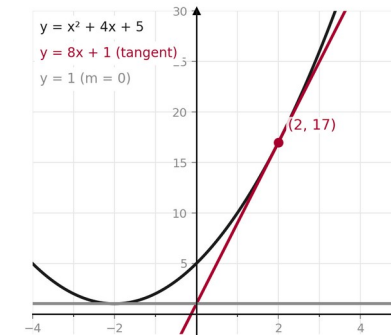
DIGITAL SAT — MATH

Advanced Math

Theory Companion — Explained Rules, Worked Problems & Cheat Sheet

Nonlinear Functions • Nonlinear Equations & Systems • Equivalent Expressions

Prepared by the CLC Digital SAT Team

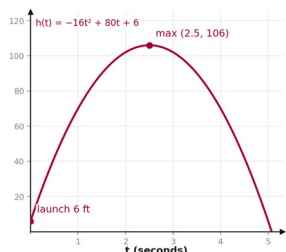


3 Skills in This Deck

Each skill opens with key terms in English and Greek, explained theory with examples, and a Watch Out slide — then 8 Hard / Very Hard worked problems. A cheat sheet closes the deck.

01

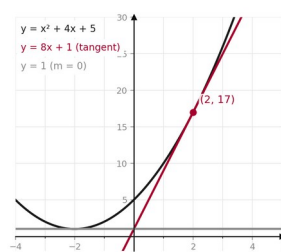
Nonlinear Functions



Quadratic and exponential functions, their graphs, models and transformations.

02

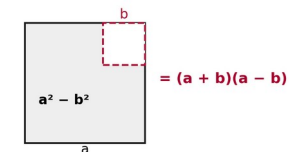
Nonlinear Equations & Systems



Quadratic, radical and rational equations; line-parabola systems.

03

Equivalent Expressions



$$= (a + b)(a - b)$$

Factoring, expanding, simplifying rational expressions and exponents.

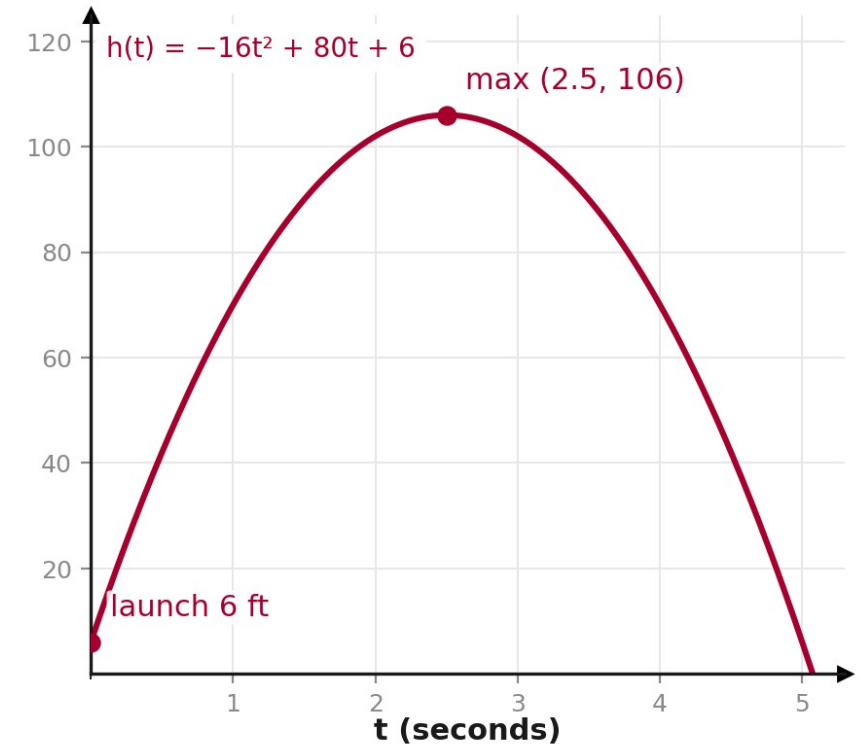
01

Nonlinear Functions

Quadratic and exponential functions, their graphs, models and transformations.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

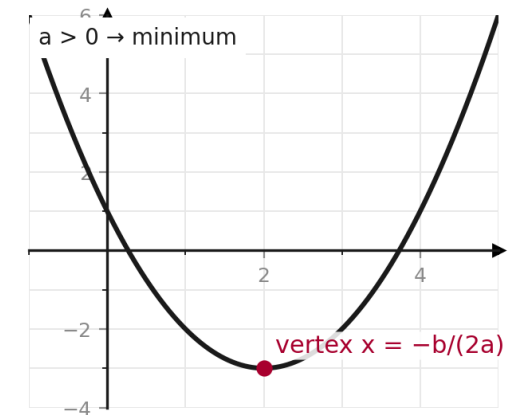


Key Terminology

Term (English)	Ελληνικά	Meaning
Quadratic function	Τετραγωνική συνάρτηση	$f(x) = ax^2 + bx + c$ or $a(x - h)^2 + k$
Parabola	Παραβολή	U-shaped graph; up if $a > 0$, down if $a < 0$
Vertex	Κορυφή	Max or min point; $x = -b/(2a)$
Axis of symmetry	Άξονας συμμετρίας	Vertical line $x = h$ through the vertex
Roots / zeros	Ρίζες	Values where $f(x) = 0$
Exponential function	Εκθετική συνάρτηση	$f(x) = a \cdot b^x$
Growth / decay factor	Συντελεστής μεταβολής	$b = 1 + r$ (growth) or $1 - r$ (decay)
Transformation	Μετασχηματισμός	Shift, stretch or reflect a graph
Multiplicity	Πολλαπλότητα	Even: touches the axis; odd: crosses it

A 文 The big picture

- About 15% of Digital SAT Math questions.
- Quadratics: vertex, zeros, max/min.
- Exponentials: initial value and growth factor, often in context.



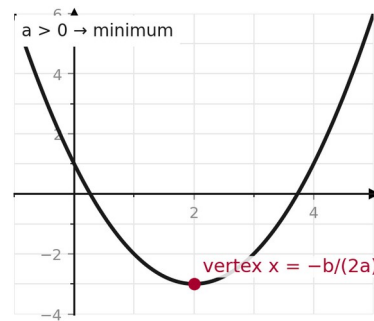
Theory Explained (1/3)

Standard Form & Vertex

$$f(x) = ax^2 + bx + c \rightarrow \text{vertex } x = -b/(2a)$$

WHAT IT MEANS

- $a > 0$ opens up (minimum); $a < 0$ opens down (maximum).
- Find the vertex x , then substitute to get the max/min value.
- c is the y -intercept — not the minimum.



Example

$$g(x) = 3x^2 - 12x + 7 \rightarrow x = 2, \text{ minimum } g(2) = -5.$$

Vertex Form

$$f(x) = a(x - h)^2 + k \rightarrow \text{vertex } (h, k)$$

WHAT IT MEANS

- h appears with a minus sign: $(x - 3)^2 \rightarrow h = +3$.
- k is the max or min value.
- Solve $f(x) = c$ by isolating $(x - h)^2$ first.



Example

$$-2(x - 4)^2 + 50: \text{ vertex } (4, 50), \text{ maximum } 50.$$

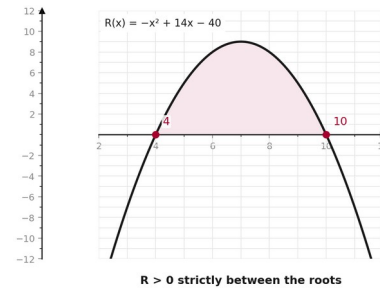
Theory Explained (2/3)

Zeros & Factored Form

$$f(x) = a(x - r_1)(x - r_2) \rightarrow \text{zeros } r_1, r_2$$

WHAT IT MEANS

- Set each factor to 0 to read the zeros.
- $(x + a)^2$ gives a double zero: the graph touches the axis.
- Downward parabola: positive strictly BETWEEN its zeros.



Example

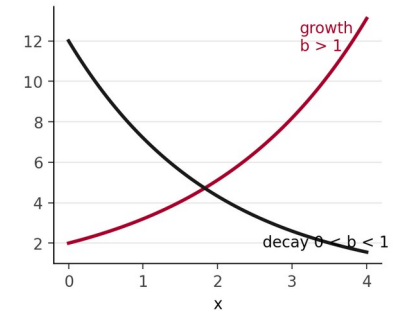
$$-(x - 4)(x - 10) > 0 \text{ for } 4 < x < 10.$$

Exponential Models

$$f(x) = a \cdot b^x = a(1 \pm r)^x$$

WHAT IT MEANS

- a = initial value, $f(0)$.
- $b > 1 \rightarrow$ growth by $r = b - 1$; $0 < b < 1 \rightarrow$ decay by $r = 1 - b$.
- Growth multiplies; linear growth adds.



Example

$$500(1.06)^t \rightarrow \text{start 500, grows 6\% per hour.}$$

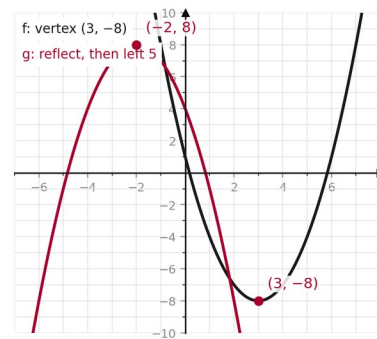
Theory Explained (3/3)

Transformations

$$f(x) + k \uparrow f(x - h) \rightarrow -f(x) \text{ reflects over x-axis}$$

WHAT IT MEANS

- Changes OUTSIDE the function move the graph vertically.
- Changes INSIDE move it horizontally — opposite to the sign.
- Track a single point (the vertex) through each step, in order.



Example

Vertex (3, -8) $\rightarrow$ reflect $\rightarrow$ (3, 8) $\rightarrow$ left 5 $\rightarrow$ (-2, 8).

Exponent Structure

$$b^{(x + c)} = b^c \cdot b^x$$

WHAT IT MEANS

- Rewrite a shifted exponent as a constant multiplier.
- Equal for all $x \implies$ matching coefficients.
- Useful for comparing models.



Example

$2^{(x + 3)} = 8 \cdot 2^x$.

Watch Out



Vertex x is $-b/(2a)$, not $-b/a$

- The missing $\div 2$ is the #1 arithmetic error.
- In $a(x - h)^2$, h is subtracted: $(x - 3)^2 \rightarrow x = +3$.



Rate $\neq$ factor

- In $a(1.08)^x$ the rate is 8%, the factor is 1.08.
- A decrease uses $(1 - r)$, never a negative base.

Problem 1

Hard

The function f is defined by $f(x) = -2(x - 4)^2 + 50$. For what value of x , where $x > 4$, does $f(x) = 18$?

- A) 7
- B) 8
- C) 9
- D) 11

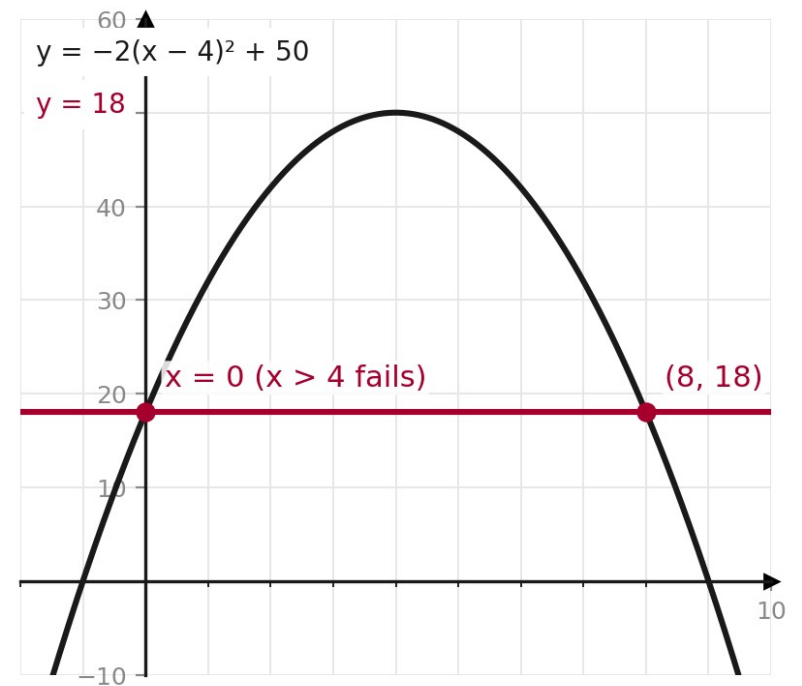
Formula / Rule

Vertex form $f(x) = a(x - h)^2 + k$: isolate the squared group, then take the square root.

Solution

1. Set the function equal to 18: $-2(x - 4)^2 + 50 = 18$.
2. Subtract 50: $-2(x - 4)^2 = -32$.
3. Divide by -2 : $(x - 4)^2 = 16$.
4. Take the square root: $x - 4 = 4$ or $x - 4 = -4$.
5. So $x = 8$ or $x = 0$.
6. Apply the condition $x > 4$: $x = 8$.

✓ **Answer: B) 8**





Problem 1 — Strategy & Elimination



Tip

Vertex form lets you skip expanding — isolate the squared group directly.



Shortcut

Subtract 50, divide by -2, take the root: $\sqrt{16} = 4$. Add to $h = 4$: $x = 8$.



Trap

Taking $x - 4 = -4$ gives $x = 0$, which ignores the condition $x > 4$. The domain restriction is deliberate.



Desmos Tip

Graph $y = -2(x-4)^2 + 50$ and $y = 18$; click the intersection with $x > 4$.



Process of Elimination

Correct: **B) 8**

Choice	Why it is right — or exactly what went wrong
A) 7	$(7 - 4)^2 = 9 \rightarrow -18 + 50 = 32 \neq 18$.
B) 8 ✓	$(8 - 4)^2 = 16 \rightarrow -32 + 50 = 18$ ✓
C) 9	$(9 - 4)^2 = 25 \rightarrow -50 + 50 = 0 \neq 18$.
D) 11	$(11 - 4)^2 = 49 \rightarrow -98 + 50 = -48 \neq 18$.

Problem 2

Hard

The function g is defined by $g(x) = 3x^2 - 12x + 7$. What is the minimum value of $g(x)$?

- A) -5
- B) -12
- C) 7
- D) 12



Formula / Rule

Vertex x-coordinate: $x = -b/(2a)$. The minimum value is g at that x .

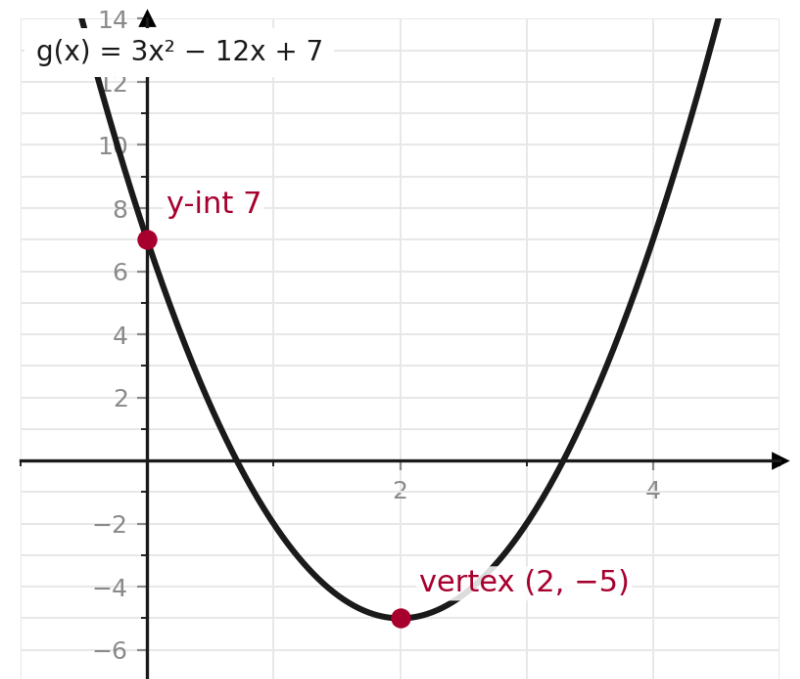


Solution

1. Identify $a = 3$ and $b = -12$.
2. Vertex: $x = -(-12)/(2 \cdot 3) = 12/6 = 2$.
3. Evaluate: $g(2) = 3(4) - 12(2) + 7$.
4. Compute: $12 - 24 + 7 = -5$.



Answer: A) -5





Problem 2 — Strategy & Elimination



Tip

$a = 3 > 0$, so the parabola opens upward and the vertex is a minimum.



Shortcut

Complete the square: $3(x - 2)^2 - 12 + 7 = 3(x - 2)^2 - 5$. The minimum is $k = -5$.



Trap

Choosing 7 — the y-intercept $g(0)$ — instead of the value at the vertex.



Desmos Tip

Graph $y = 3x^2 - 12x + 7$ and click the vertex: $(2, -5)$.



Process of Elimination

Correct: **A) -5**

Choice	Why it is right — or exactly what went wrong
A) -5 ✓	$g(2) = 12 - 24 + 7 = -5$ ✓
B) -12	Dropped the +7: $12 - 24 = -12$.
C) 7	The y-intercept $g(0)$, not the minimum.
D) 12	Computed only $3(2^2) = 12$ and stopped.

Problem 3

Hard

A population of bacteria is modeled by $P(t) = 500(1.06)^t$, where t is time in hours. Which best describes the meaning of 1.06 in this context?

- A) The initial population of bacteria
- B) The number of bacteria added each hour
- C) The population grows by 6% each hour
- D) The population doubles every 1.06 hours

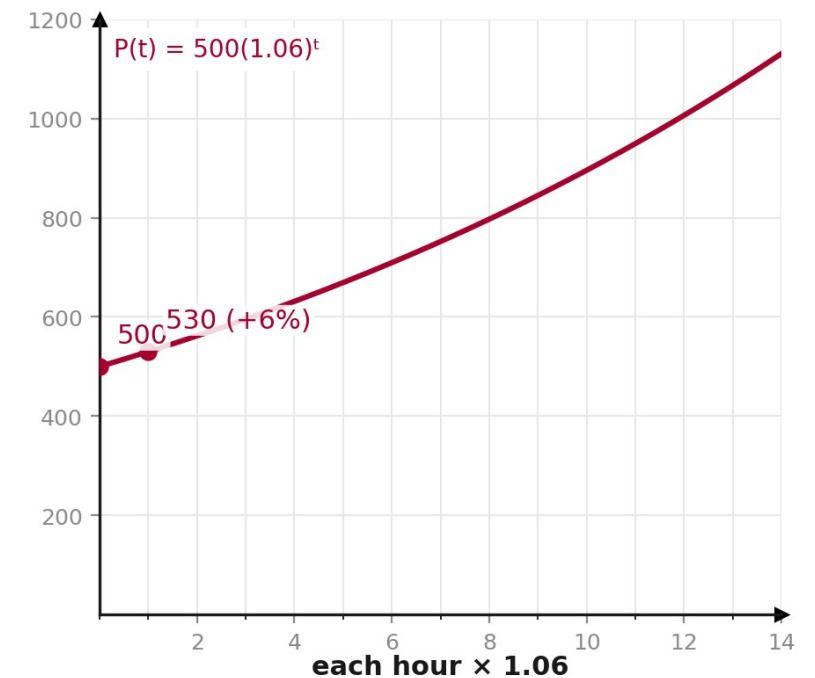
Formula / Rule

Exponential model $f(x) = a(1 + r)^x$: the base $1 + r$ means growth of r per period.

Solution

- The model has the form $a \cdot b^t$ with $a = 500$ and $b = 1.06$.
- Write the base as $1 + r$: $1.06 = 1 + 0.06$.
- So $r = 0.06 = 6\%$.
- Each hour the population is multiplied by 1.06 — a 6% increase.

✓ **Answer:** C) The population grows by 6% each hour





Problem 3 — Strategy & Elimination



Tip

Subtract 1 from the base to get the rate: $1.06 - 1 = 0.06 = 6\%$.



Shortcut

Base $> 1 \rightarrow$ growth; base $- 1 =$ rate. Read it straight from the model.



Trap

Choice B describes linear growth (adding a fixed amount). Exponential growth multiplies by a factor.



Desmos Tip

Graph $P(t) = 500(1.06)^t$: $P(0) = 500$, $P(1) = 530$ — a 6% increase.



Process of Elimination

Correct: C) The population grows by 6% each hour

Choice	Why it is right — or exactly what went wrong
A) Initial population	The initial value is $a = 500$, not 1.06.
B) Bacteria added each hour	That is linear growth; exponential growth multiplies.
C) Grows 6% each hour ✓	$1.06 = 1 + 0.06 \rightarrow 6\%$ growth per hour ✓
D) Doubles every 1.06 h	Doubling needs $1.06^t = 2 \rightarrow t \approx 12$ hours.

Problem 4

Very Hard

The graph of $y = f(x)$ is a parabola with vertex $(3, -8)$. The graph of $y = g(x)$ is obtained by reflecting $y = f(x)$ over the x -axis and then shifting it 5 units to the left. What is the vertex of $y = g(x)$?

- A) $(-2, 8)$
- B) $(-2, -8)$
- C) $(8, 8)$
- D) $(-8, 8)$

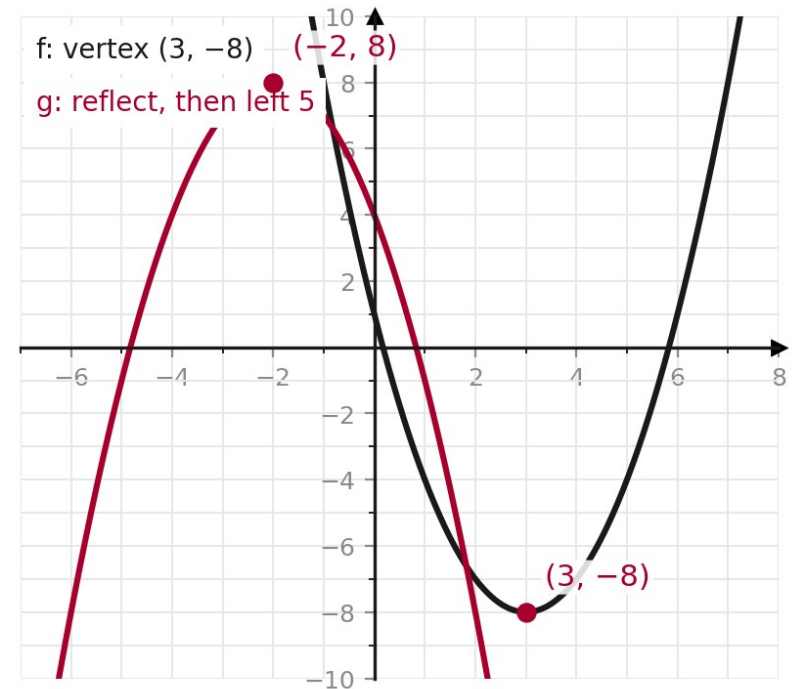
Formula / Rule

Reflect over the x -axis: $(x, y) \rightarrow (x, -y)$. Shift left 5: $(x, y) \rightarrow (x - 5, y)$.

Solution

1. Start with the vertex $(3, -8)$.
2. Reflect over the x -axis: the y -coordinate changes sign $\rightarrow (3, 8)$.
3. Shift 5 units left: subtract 5 from $x \rightarrow (3 - 5, 8)$.
4. Vertex of g : $(-2, 8)$.

✓ **Answer:** A) $(-2, 8)$





Problem 4 — Strategy & Elimination



Tip

Apply transformations one at a time, in the stated order. Track only the vertex point.



Shortcut

$(3, -8) \rightarrow \text{reflect} \rightarrow (3, 8) \rightarrow \text{left } 5 \rightarrow (-2, 8)$. No algebra needed.



Trap

Forgetting to flip the y-coordinate, or shifting right instead of left.



Desmos Tip

Graph $f(x) = (x-3)^2 - 8$ and $g(x) = -f(x+5)$; click the vertex of g.



Process of Elimination

Correct: **A) $(-2, 8)$**

Choice	Why it is right — or exactly what went wrong
A) $(-2, 8)$ ✓	Reflect: $-8 \rightarrow 8$. Shift left 5: $3 \rightarrow -2$ ✓
B) $(-2, -8)$	Forgot to flip y when reflecting over the x-axis.
C) $(8, 8)$	Shifted right instead of left: $3 + 5 = 8$.
D) $(-8, 8)$	Mixed up the coordinates after reflecting.

Problem 5

Hard

For $f(x) = x^2 + 6x + 9$, which equivalent form most directly reveals the zeros of f ?

- A) $f(x) = (x + 3)^2$
- B) $f(x) = (x + 9)(x - 1)$
- C) $f(x) = x(x + 6) + 9$
- D) $f(x) = (x + 3)(x - 3) + 18$

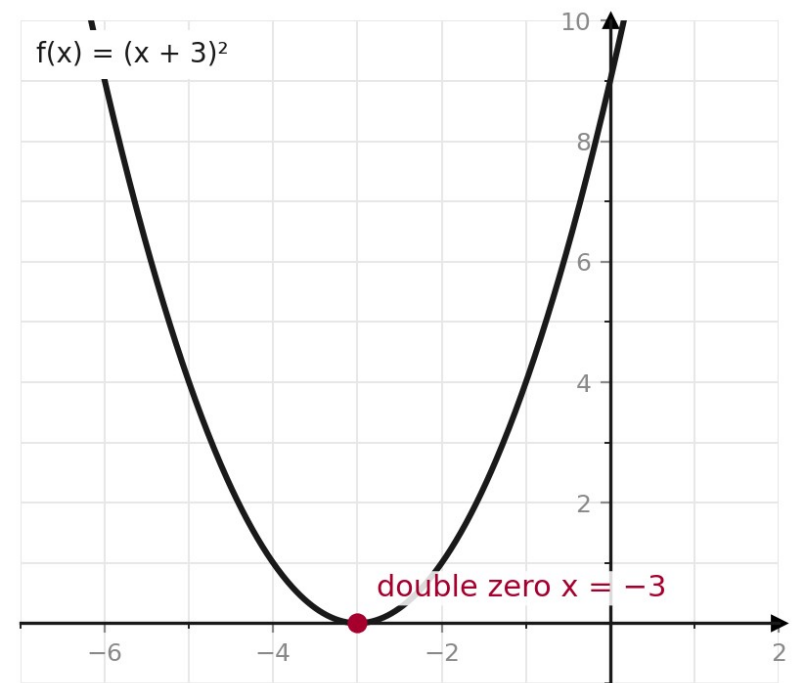
Formula / Rule

Factored form shows the zeros: set each factor to 0. $(x + a)^2$ gives a double zero at $x = -a$.

Solution

1. Check the pattern $a^2 + 2ab + b^2$: $x^2 + 2(3)x + 3^2$.
2. So $x^2 + 6x + 9 = (x + 3)^2$.
3. Set $(x + 3)^2 = 0 \rightarrow x = -3$.
4. The zero $x = -3$ is visible immediately — a double zero.

✓ **Answer:** A) $f(x) = (x + 3)^2$





Problem 5 — Strategy & Elimination



Tip

'Most directly reveals the zeros' means factored form — no further algebra needed.



Shortcut

Middle term $2 \cdot 3 = 6$ and last term $3^2 = 9 \rightarrow$ perfect square trinomial.



Trap

Choice C is equivalent but still needs algebra to find the zero.



Desmos Tip

Graph $y = x^2 + 6x + 9$: it touches the x-axis only at $x = -3$.



Process of Elimination

Correct: **A) $f(x) = (x + 3)^2$**

Choice	Why it is right — or exactly what went wrong
A) $(x + 3)^2$ ✓	Set to 0: $x = -3$ immediately ✓
B) $(x + 9)(x - 1)$	Expands to $x^2 + 8x - 9$ — not equivalent.
C) $x(x + 6) + 9$	Equivalent, but the zero is not visible.
D) $(x + 3)(x - 3) + 18$	$= x^2 + 9$ — not equivalent.

Problem 6

Hard

The height, in feet, of a projectile t seconds after launch is $h(t) = -16t^2 + 80t + 6$. What is the maximum height, in feet?

- A) 80
- B) 100
- C) 106
- D) 112

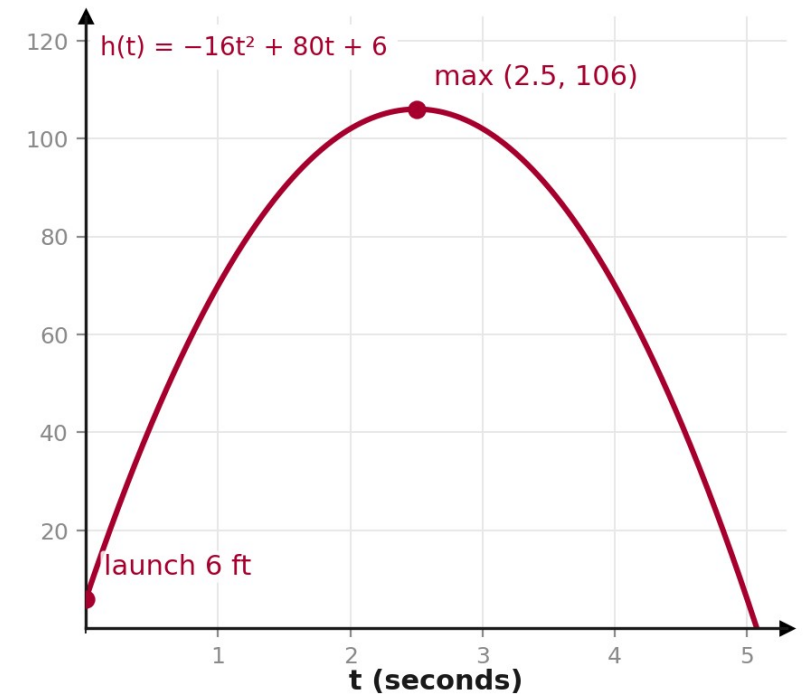
Formula / Rule

Vertex: $t = -b/(2a)$. Maximum height = h at that t .

Solution

1. Identify $a = -16$ and $b = 80$.
2. Vertex time: $t = -80/(2 \cdot -16) = 2.5$ seconds.
3. Evaluate: $h(2.5) = -16(2.5^2) + 80(2.5) + 6$.
4. Compute: $-100 + 200 + 6 = 106$ feet.

✓ **Answer: C) 106**





Problem 6 — Strategy & Elimination



Tip

$a < 0$, so the parabola opens downward and the vertex is the maximum.



Shortcut

$t = 2.5$, then evaluate the full expression — including the constant.



Trap

Leaving out the +6 launch height gives 100.



Desmos Tip

Graph $y = -16x^2 + 80x + 6$; the peak is labeled (2.5, 106).



Process of Elimination

Correct: **C) 106**

Choice	Why it is right — or exactly what went wrong
A) 80	The coefficient 80 (initial velocity), not a height.
B) 100	Omitted the +6 launch height.
C) 106 ✓	$-100 + 200 + 6 = 106$ ✓
D) 112	Added the 6 twice.

Problem 7

Very Hard

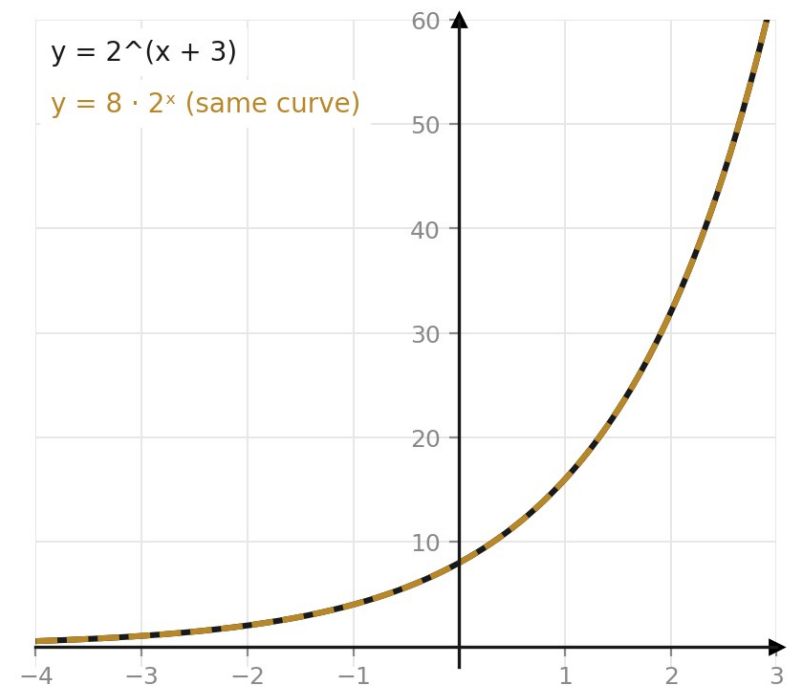
The function f is defined by $f(x) = 2^{(x+3)}$. The function g is defined by $g(x) = k \cdot 2^x$, where k is a constant. If $g(x) = f(x)$ for all x , what is the value of k ?
(Student-produced response — enter your answer.)

**Formula / Rule**

Exponent rule: $2^{(a+b)} = 2^a \cdot 2^b$.

**Solution**

1. Split the exponent: $2^{(x+3)} = 2^x \cdot 2^3$.
2. Evaluate $2^3 = 8$, so $f(x) = 8 \cdot 2^x$.
3. Compare with $g(x) = k \cdot 2^x$.
4. The coefficients must match: $k = 8$.

**Answer: 8**



Problem 7 — Strategy & Elimination



Tip

When two exponential expressions with the same base are equal for all x , rewrite and match coefficients.



Shortcut

$2^{(x+3)} = 8 \cdot 2^x$ by inspection, so $k = 8$.



Trap

Writing 2^3 as 6 (2×3) instead of 8 is a frequent slip.



Desmos Tip

Graph $y = 2^{(x+3)}$ and $y = 8 \cdot 2^x$: the curves coincide.



Process of Elimination

Correct: 8

Tempting wrong entry	Exactly what went wrong
3	Took the exponent 3 as the coefficient.
6	Computed 2×3 instead of 2^3 .
16	Used 2^4 — miscounted the exponent.

Problem 8

Very Hard

A company's revenue R , in thousands of dollars, x years after launch is modeled by $R(x) = -x^2 + 14x - 40$. In which years after launch is the revenue positive ($R > 0$)?

- A) Years 1 through 5
- B) Years 4 through 10
- C) Years 5 through 9
- D) Years 6 through 8

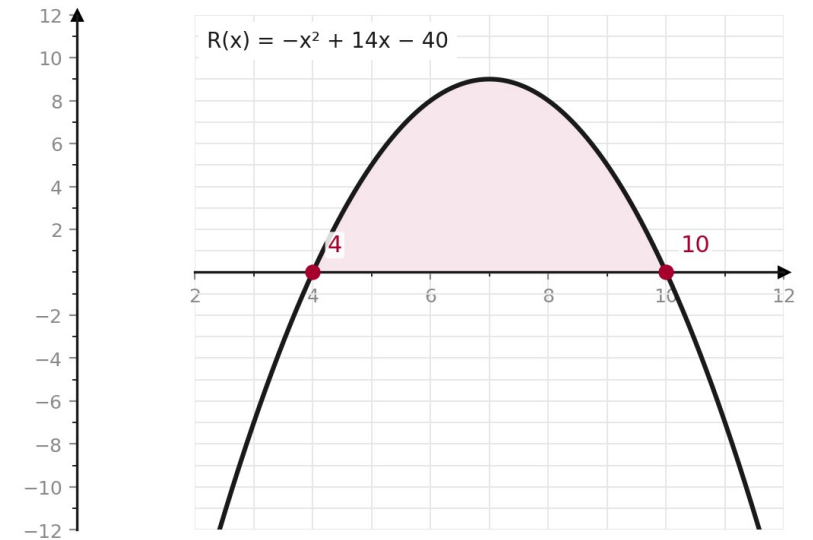
Formula / Rule

Find the roots; a downward parabola is positive strictly BETWEEN its roots.

Solution

- Factor out -1 : $R(x) = -(x^2 - 14x + 40)$.
- Factor: $R(x) = -(x - 4)(x - 10)$.
- Roots: $x = 4$ and $x = 10$, where $R = 0$.
- The parabola opens down, so $R > 0$ for $4 < x < 10$.
- Whole years strictly between the roots: 5, 6, 7, 8, 9.

✓ **Answer:** C) Years 5 through 9



$R > 0$ strictly between the roots



Problem 8 — Strategy & Elimination



Tip

A downward parabola is positive strictly between its roots — not at the roots.



Shortcut

Roots 4 and 10 → positive for $4 < x < 10$ → years 5 to 9.



Trap

Including years 4 and 10, where revenue is exactly 0 (not positive).



Desmos Tip

Graph $y = -x^2 + 14x - 40$: above the x-axis between $x = 4$ and $x = 10$.



Process of Elimination

Correct: C) Years 5 through 9

Choice	Why it is right — or exactly what went wrong
A) Years 1–5	$R(1) = -27 < 0$ — wrong interval.
B) Years 4–10	Includes 4 and 10, where $R = 0$.
C) Years 5–9 ✓	$R(5) = 5 > 0$ and $R(9) = 5 > 0$ ✓
D) Years 6–8	Too narrow — years 5 and 9 are also positive.

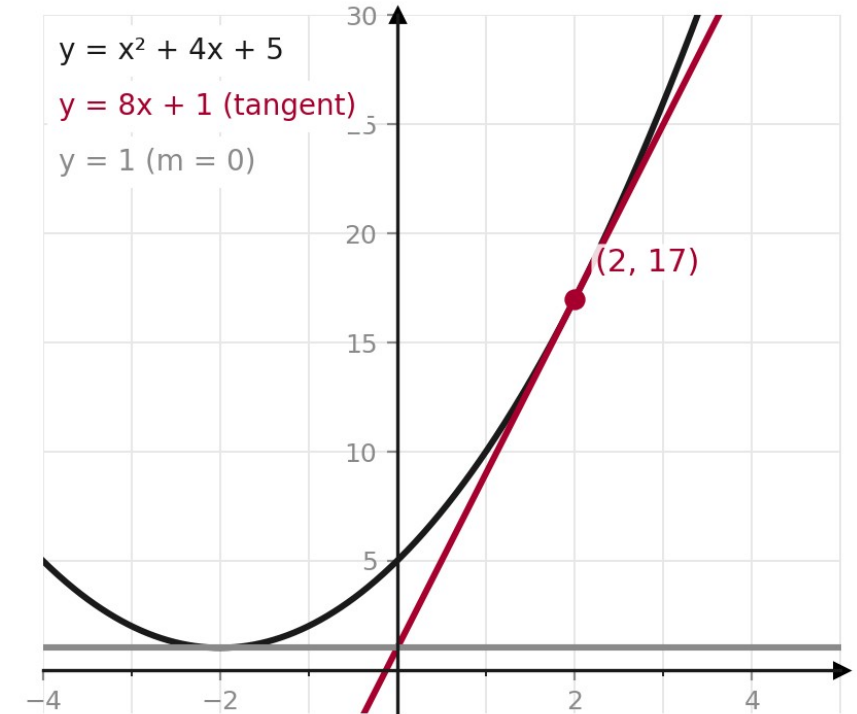
02

Nonlinear Equations & Systems

Quadratic, radical and rational equations; line–parabola systems.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

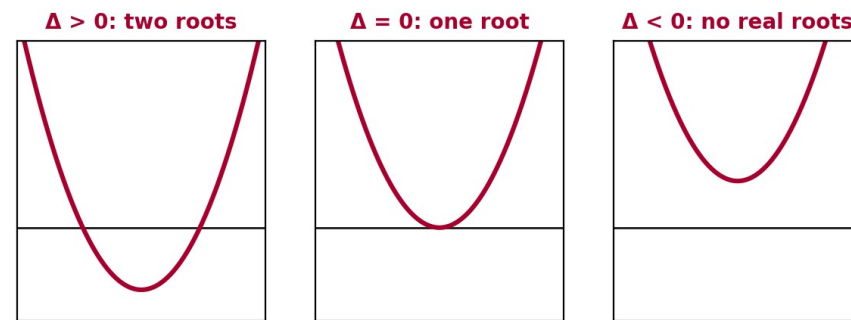


Key Terminology

Term (English)	Ελληνικά	Meaning
Quadratic equation	Τετραγωνική εξίσωση	Degree 2; at most 2 real solutions
Factoring	Παραγοντοποίηση	Rewrite as a product
Completing the square	Συμπλήρωση τετραγώνου	Create a perfect-square trinomial
Quadratic formula	Τύπος δευτεροβάθμιας	$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$
Discriminant	Διακρίνουσα	$\Delta = b^2 - 4ac$; counts real roots
Extraneous solution	Ξένη λύση	Solves a transformed equation, not the original
System of equations	Σύστημα εξισώσεων	Line and parabola meet 0, 1 or 2 times
Vieta's formulas	Τύποι Vieta	Sum = $-b/a$, product = c/a
Radical equation	Άρρητη εξίσωση	Variable under a root sign

A 支 The big picture

- About 12% of Digital SAT Math questions.
- Often asks HOW MANY solutions — not what they are.
- Extraneous roots are deliberate traps.



Theory Explained (1/3)

Solving Quadratics

factor · complete the square · $x = (-b \pm \sqrt{\Delta}) / (2a)$

WHAT IT MEANS

- Factor when roots are 'nice' (ac method when $a \neq 1$).
- The quadratic formula always works.
- Apply any extra condition (like $x > 0$) at the end.



Example

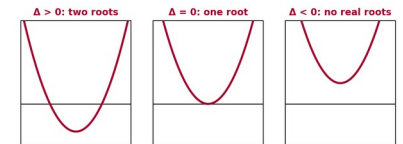
$$3x^2 + 5x - 2 = (3x - 1)(x + 2) \rightarrow x = 1/3 \text{ or } -2.$$

The Discriminant

$\Delta = b^2 - 4ac$: > 0 two · $= 0$ one · < 0 none

WHAT IT MEANS

- Counts real solutions without solving.
- $\Delta = 0$ means the graph is tangent to the x-axis.
- 'Exactly one' $\rightarrow \Delta = 0$; 'two' $\rightarrow \Delta > 0$.



Example

$$x^2 - 10x + k: 100 - 4k = 0 \rightarrow k = 25.$$

Theory Explained (2/3)

Vieta's Formulas

$$r_1 + r_2 = -b/a \quad r_1 \cdot r_2 = c/a$$

WHAT IT MEANS

- Gives sum or product without finding the roots.
- Always divide by a — the leading coefficient.
- Write the equation in standard form first.



Example

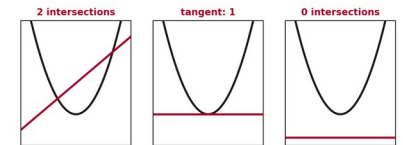
$2x^2 - 14x + 9$: sum = $14/2 = 7$, product = $9/2$.

Line-Parabola Systems

substitute → one quadratic → check Δ

WHAT IT MEANS

- Set the y -expressions equal and move everything to one side.
- Δ of the combined quadratic = number of intersections.
- Tangent line $\iff \Delta = 0$.



Example

$x^2 + 4x + 5 = mx + 1 \rightarrow \Delta = (4 - m)^2 - 16 = 0$.

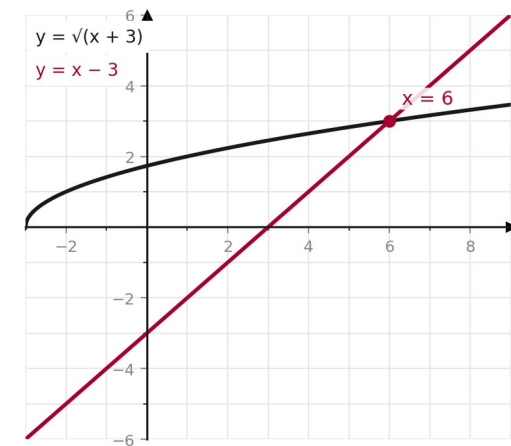
Theory Explained (3/3)

Radical Equations

isolate → square → solve → CHECK

WHAT IT MEANS

- Squaring can create extraneous roots.
- A square root is never negative: the other side must be ≥ 0 .
- Test every candidate in the ORIGINAL equation.



$x = 1$ is extraneous: $\sqrt{4} \neq -2$



Example

$$\sqrt{x+3} = x-3 \rightarrow x=6 \checkmark, x=1 \times.$$

Watch Out



Check in the ORIGINAL equation

- After squaring or clearing denominators, some roots are fake.
- Extraneous roots appear as answer choices on purpose.



$\Delta = 0$ means ONE solution

- Not zero solutions — a repeated root.
- Boundary values of k give $\Delta = 0$, not two solutions.

Problem 1

Hard

For what value of x is $3x^2 + 5x - 2 = 0$ and $x > 0$?

- A) $1/3$
- B) $2/3$
- C) 1
- D) 2

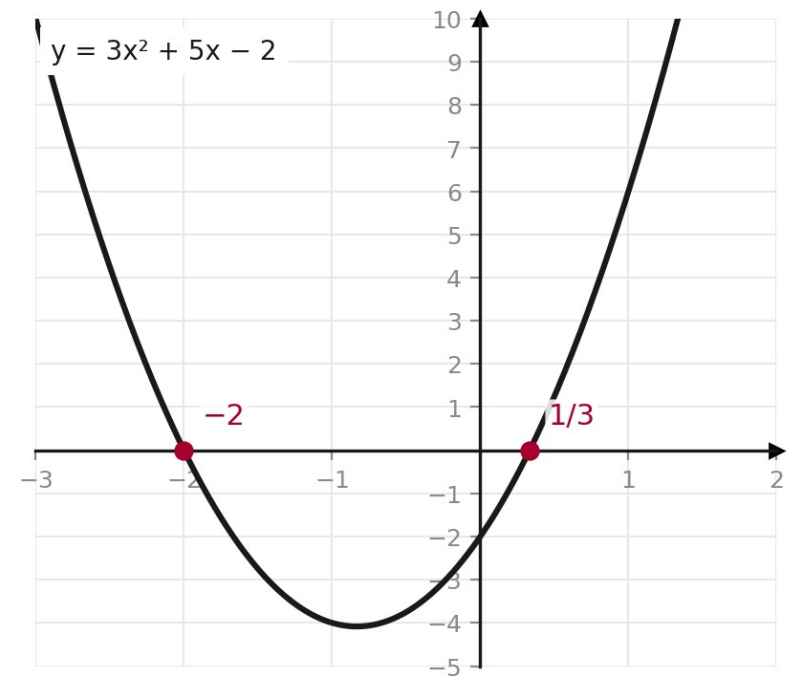
Formula / Rule

Factor (ac method) or use the quadratic formula; keep the positive root.

Solution

1. Find two numbers with product $a \cdot c = -6$ and sum $b = 5$: 6 and -1 .
2. Factor: $(3x - 1)(x + 2) = 0$.
3. Roots: $x = 1/3$ or $x = -2$.
4. Apply $x > 0$: $x = 1/3$.

 **Answer:** A) $1/3$





Problem 1 — Strategy & Elimination



Tip

With $a \neq 1$, look for two numbers whose product is $a \cdot c$ and sum is b .



Trap

Stopping at $x = -2$ or choosing a value that is not a root at all.



Shortcut

Test the choices: $1/3 + 5/3 - 2 = 0$. Back-substitution beats factoring.



Desmos Tip

Graph $y = 3x^2 + 5x - 2$; the positive x -intercept is $1/3 \approx 0.333$.



Process of Elimination

Correct: **A) 1/3**

Choice	Why it is right — or exactly what went wrong
A) 1/3 ✓	$3(1/9) + 5(1/3) - 2 = 0$ ✓
B) 2/3	$4/3 + 10/3 - 2 = 8/3 \neq 0$.
C) 1	$3 + 5 - 2 = 6 \neq 0$.
D) 2	$12 + 10 - 2 = 20 \neq 0$.

Problem 2

Hard

If $x^2 - 10x + k = 0$ has exactly one real solution, what is the value of k ?

- A) 5
- B) 10
- C) 25
- D) 100

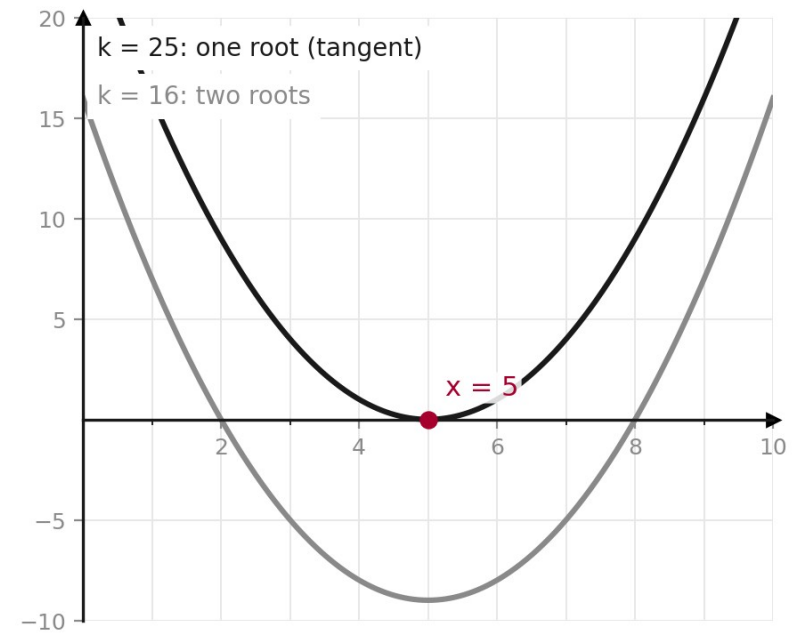
Formula / Rule

Exactly one real solution $\iff$ discriminant $\Delta = b^2 - 4ac = 0$.

Solution

1. Identify $a = 1$, $b = -10$, $c = k$.
2. Set $\Delta = 0$: $(-10)^2 - 4(1)(k) = 0$.
3. Simplify: $100 - 4k = 0$.
4. Solve: $k = 25$.

Answer: C) 25





Problem 2 — Strategy & Elimination



Tip

One real solution means the parabola is tangent to the x-axis → $\Delta = 0$.



Shortcut

With $a = 1$, one root when $c = (b/2)^2 = (-5)^2 = 25$.



Trap

Stopping at $b^2 = 100$ and forgetting to divide by 4.



Desmos Tip

Graph $y = x^2 - 10x + 25$: it touches the x-axis only at $x = 5$.



Process of Elimination

Correct: **C) 25**

Choice	Why it is right — or exactly what went wrong
A) 5	$\Delta = 100 - 20 = 80 > 0 \rightarrow$ two roots.
B) 10	$\Delta = 100 - 40 = 60 > 0 \rightarrow$ two roots.
C) 25 ✓	$\Delta = 100 - 100 = 0 \rightarrow$ one root ✓
D) 100	$\Delta = 100 - 400 < 0 \rightarrow$ no real roots.

Problem 3

Hard

The sum of the roots of $2x^2 - 14x + 9 = 0$ is equal to which of the following?

- A) $7/2$
- B) 7
- C) $9/2$
- D) 14

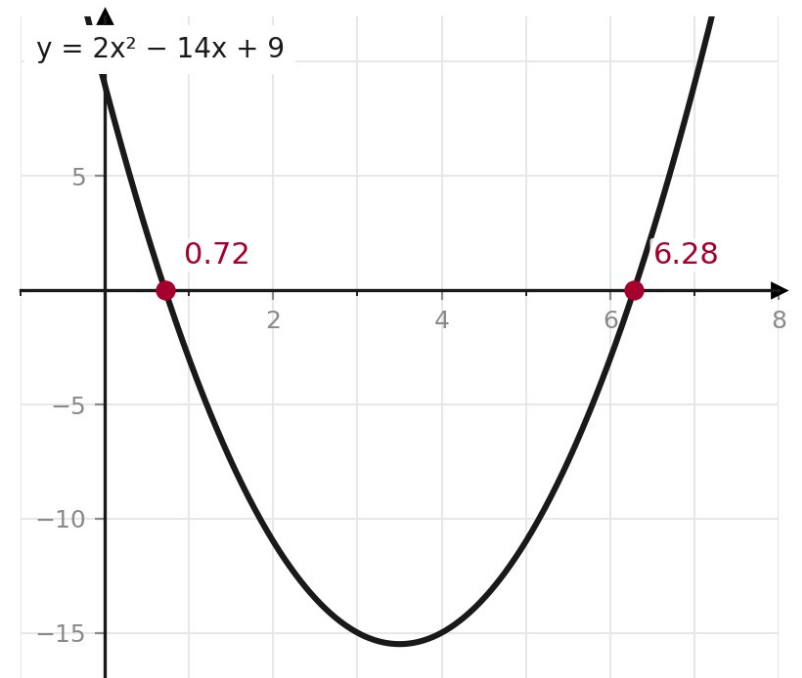
Formula / Rule

Vieta: sum of roots = $-b/a$ (divide by the leading coefficient).

Solution

1. Identify $a = 2$ and $b = -14$.
2. Sum of roots = $-b/a = -(-14)/2$.
3. Simplify: $14/2 = 7$.

✓ **Answer: B) 7**



$$0.72 + 6.28 = 7$$



Problem 3 — Strategy & Elimination



Tip

When $a \neq 1$, the sum is $-b/a$ — not $-b$.



Shortcut

Sum = $-b/a = 14/2 = 7$ — no need to find the roots.



Trap

Reporting 14 (forgetting to divide by a), or dividing by $2a$ instead of a .



Desmos Tip

Find both x -intercepts of $y = 2x^2 - 14x + 9$ and add them: $\approx 0.72 + 6.28 = 7$.



Process of Elimination

Correct: **B) 7**

Choice	Why it is right — or exactly what went wrong
A) $7/2$	Divided by $2a = 4$ instead of $a = 2$.
B) 7 ✓	$-(-14)/2 = 7$ ✓
C) $9/2$	That is c/a — the PRODUCT of the roots.
D) 14	$-b$ alone; forgot to divide by $a = 2$.

Problem 4

Very Hard

The system $y = x^2 - 5x + 6$ and $y = 3x - 9$ has two solutions. What is the sum of the x-values of the solutions?

- A) 2
- B) 4
- C) 8
- D) 15

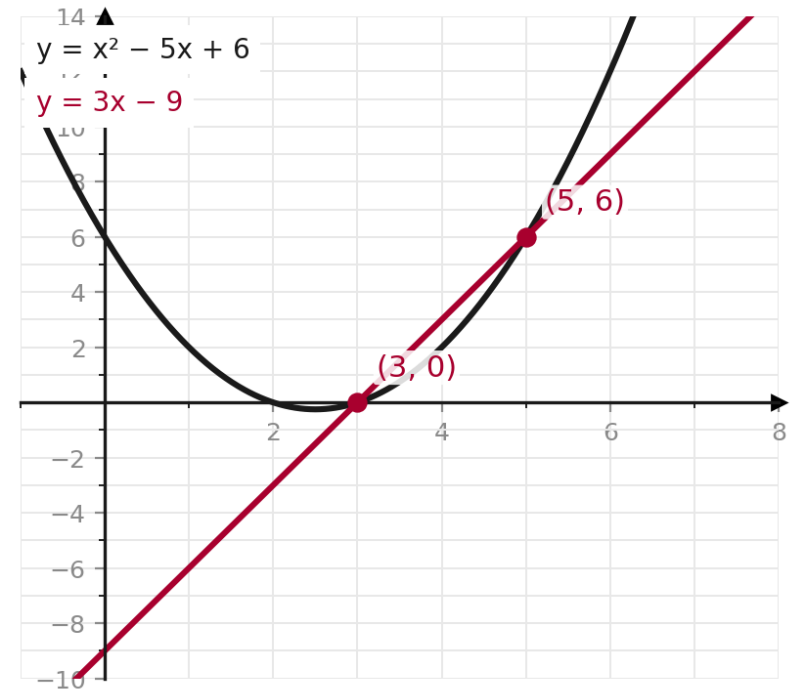
Formula / Rule

Substitute the line into the quadratic; then use Vieta: $\text{sum} = -b/a$.

Solution

1. Set the expressions equal: $x^2 - 5x + 6 = 3x - 9$.
2. Move every term left: $x^2 - 8x + 15 = 0$.
3. Sum of roots = $-(-8)/1 = 8$.
4. Check: roots are 3 and 5; $3 + 5 = 8$.

✓ **Answer: C) 8**





Problem 4 — Strategy & Elimination



Tip

You only need the sum — Vieta on the combined quadratic saves time.



Shortcut

Substitute $\rightarrow x^2 - 8x + 15 = 0 \rightarrow \text{sum} = 8$.



Trap

Moving $3x$ with the wrong sign produces $x^2 - 2x + 15 = 0$ and a sum of 2.



Desmos Tip

Graph both equations; click the two intersections $(3, 0)$ and $(5, 6)$.



Process of Elimination

Correct: **C) 8**

Choice	Why it is right — or exactly what went wrong
A) 2	Sign slip: wrote $x^2 - 2x + 15 = 0$.
B) 4	Halved the sum (the average of the roots).
C) 8 ✓	$x^2 - 8x + 15 = 0 \rightarrow \text{sum} = 8$ ✓
D) 15	That is the PRODUCT of the roots (c/a) .

Problem 5

Hard

What is the solution set of $\sqrt{x+3} = x-3$?

- A) {6}
- B) {1}
- C) {1, 6}
- D) {-1, 6}

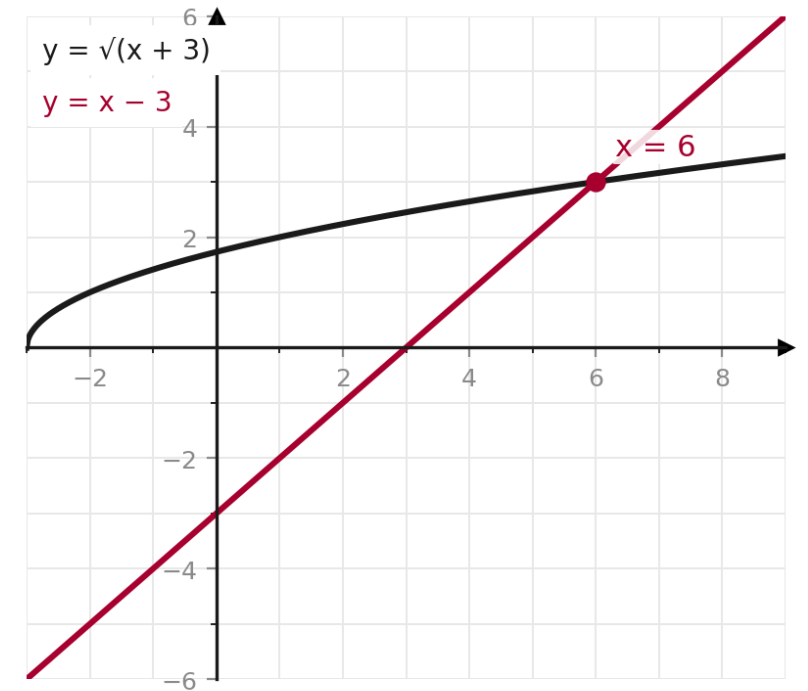
Formula / Rule

Isolate the radical, square both sides, solve, then CHECK every candidate.

Solution

1. The radical is already isolated. Square both sides: $x+3 = (x-3)^2$.
2. Expand: $x+3 = x^2 - 6x + 9$.
3. Rearrange: $x^2 - 7x + 6 = 0$.
4. Factor: $(x-1)(x-6) = 0 \rightarrow x = 1$ or $x = 6$.
5. Check $x = 6$: $\sqrt{9} = 3$ and $6 - 3 = 3$ ✓
6. Check $x = 1$: $\sqrt{4} = 2$ but $1 - 3 = -2$ ✗ (extraneous).
7. Solution set: {6}.

✓ **Answer:** A) {6}



x = 1 is extraneous: $\sqrt{4} \neq -2$



Problem 5 — Strategy & Elimination



Tip

Always substitute candidates into the ORIGINAL equation, not the squared one.



Shortcut

A square root is never negative, so $x - 3 \geq 0 \rightarrow x \geq 3$. That rules out $x = 1$ at once.



Trap

Reporting both roots of the squared equation — $x = 1$ is extraneous.



Desmos Tip

Graph $y = \sqrt{x+3}$ and $y = x - 3$; they meet only at $x = 6$.



Process of Elimination

Correct: A) {6}

Choice	Why it is right — or exactly what went wrong
A) {6} ✓	Only $x = 6$ satisfies the original equation ✓
B) {1}	Kept the extraneous root and dropped the real one.
C) {1, 6}	Never checked the roots in the original equation.
D) {-1, 6}	Sign error when factoring: $(x + 1)(x - 6)$.

Problem 6

Hard

The equation $2x^2 + kx + 8 = 0$ has two real solutions. Which of the following could be the value of k ?

- A) -9
- B) -8
- C) 0
- D) 8

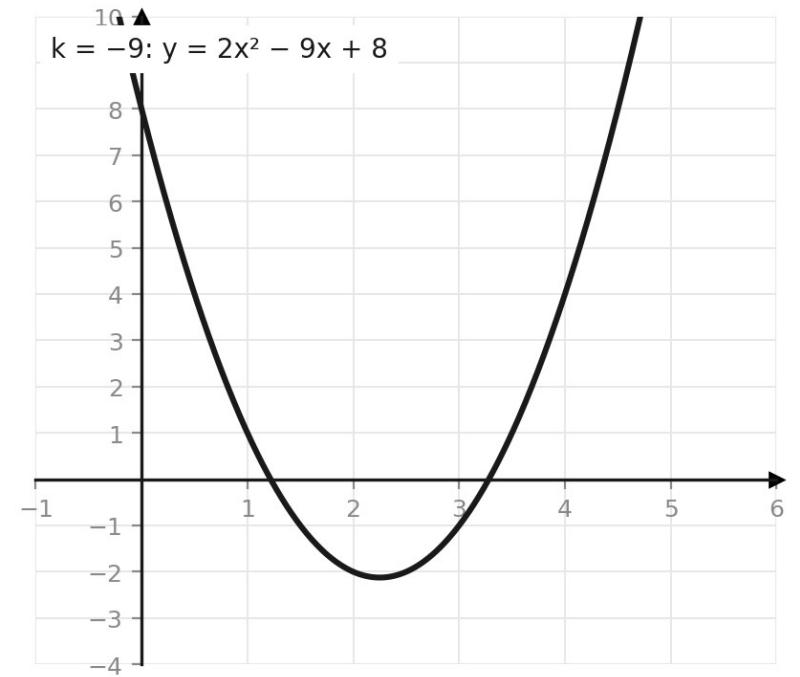
Formula / Rule

Two real solutions requires $\Delta = b^2 - 4ac > 0$.

Solution

1. Compute $\Delta = k^2 - 4(2)(8) = k^2 - 64$.
2. Two real solutions: $k^2 - 64 > 0$.
3. So $k^2 > 64$, which means $|k| > 8$.
4. Only $k = -9$ satisfies $|k| > 8$.

✓ **Answer:** A) -9



$\Delta = 17 > 0 \rightarrow$ two x-intercepts



Problem 6 — Strategy & Elimination



Tip

Set $\Delta > 0$, solve for k , then test the choices.



Shortcut

$|k| > 8$ — look for the only choice outside -8 to 8 .



Trap

$k = \pm 8$ gives $\Delta = 0$ — exactly ONE solution. The boundary is a deliberate trap.



Desmos Tip

With $k = -9$, graph $y = 2x^2 - 9x + 8$: it crosses the x -axis twice.



Process of Elimination

Correct: **A) -9**

Choice	Why it is right — or exactly what went wrong
A) -9 ✓	$\Delta = 81 - 64 = 17 > 0 \rightarrow$ two solutions ✓
B) -8	$\Delta = 64 - 64 = 0 \rightarrow$ only one solution.
C) 0	$\Delta = -64 < 0 \rightarrow$ no real solutions.
D) 8	$\Delta = 0 \rightarrow$ only one solution.

Problem 7

Very Hard

The system $y = x^2 + 4x + 5$ and $y = mx + 1$ has exactly one solution, where m is a positive constant. What is the value of m ?
(Student-produced response — enter your answer.)

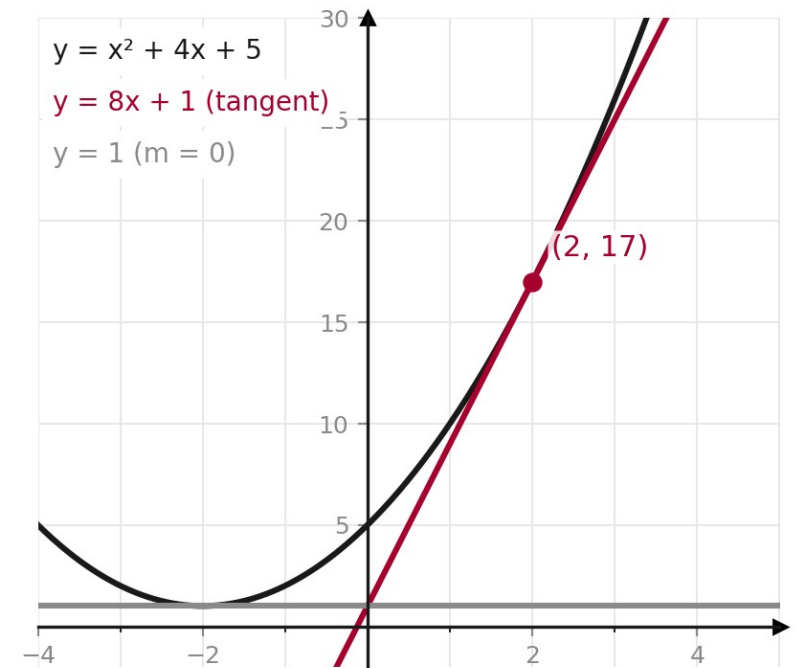
Formula / Rule

Substitute the line into the parabola; exactly one solution (tangency) $\iff \Delta = 0$.

Solution

1. Set equal: $x^2 + 4x + 5 = mx + 1$.
2. Rearrange: $x^2 + (4 - m)x + 4 = 0$.
3. Tangency: $\Delta = (4 - m)^2 - 4(1)(4) = 0$.
4. So $(4 - m)^2 = 16 \rightarrow 4 - m = 4$ or $4 - m = -4$.
5. $m = 0$ or $m = 8$.
6. m must be positive: $m = 8$.

✓ **Answer: 8**





Problem 7 — Strategy & Elimination



Tip

Line tangent to a parabola $\iff \Delta = 0$ for the combined quadratic.



Shortcut

$(4 - m)^2 = 16 \rightarrow 4 - m = \pm 4 \rightarrow m = 0$ or $8 \rightarrow$ positive answer 8.



Trap

Solving $(4 - m)^2 = 16$ with only one sign, or reporting $m = 0$, which the condition $m > 0$ excludes.



Desmos Tip

Graph $y = x^2 + 4x + 5$ and $y = 8x + 1$: the line touches the parabola once, at $(2, 17)$.



Process of Elimination

Correct: 8

Tempting wrong entry	Exactly what went wrong
0	The other tangent line — excluded because m must be positive.
4	Set $4 - m = 0$: then $\Delta = -16 < 0$ (no solution).
-8	Sign error: $4 - m = -4$ gives $m = 8$, not -8 .

Problem 8

Very Hard

The product of the two solutions of $3x^2 - 15x + c = 0$ is 8.
What is the value of c ?

- A) 8
- B) 15
- C) 24
- D) 40

Formula / Rule

Vieta: product of roots = c/a .

First move

Vieta's product is c/a , not c . When $a \neq 1$, multiply by a .

Solution

1. Identify $a = 3$.
2. Product of roots = $c/3$.
3. Set $c/3 = 8$.
4. Solve: $c = 24$.

 **Answer:** C) 24

Quick check

$c = \text{product} \times a = 8 \times 3 = 24$.



Problem 8 — Strategy & Elimination



Tip

Vieta's product is c/a , not c . When $a \neq 1$, multiply by a .



Trap

Answering 8 — the product itself — without multiplying by $a = 3$.



Shortcut

$c = \text{product} \times a = 8 \times 3 = 24$.



Desmos Tip

Graph $y = 3x^2 - 15x + 24$, click both x-intercepts, and multiply them: the product is 8.



Process of Elimination

Correct: **C) 24**

Choice	Why it is right — or exactly what went wrong
A) 8	The product itself; forgot to multiply by $a = 3$.
B) 15	Used the b-coefficient — confused sum with product.
C) 24 ✓	$c/3 = 8 \rightarrow c = 24$ ✓
D) 40	Multiplied 8×5 instead of 8×3 .

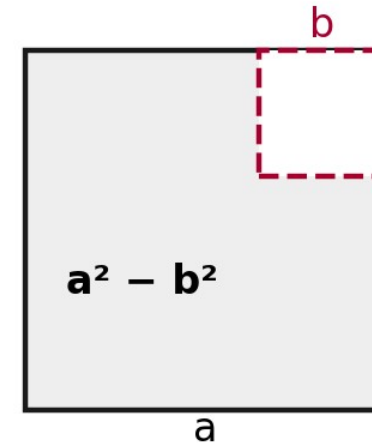
03

Equivalent Expressions

Factoring, expanding, simplifying rational expressions and exponents.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



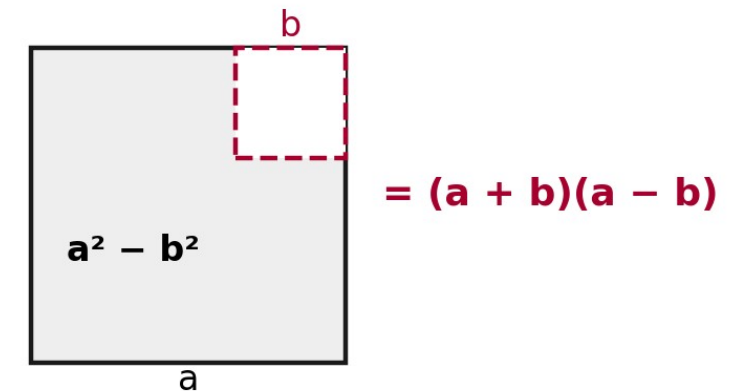
$$= (a + b)(a - b)$$

Key Terminology

Term (English)	Ελληνικά	Meaning
Polynomial	Πολυώνυμο	Sum of terms ax^n , classified by degree
Like terms	Όμοιοι όροι	Same variable part; can be combined
GCF	Μέγιστος κοινός διαιρέτης	Largest factor common to all terms
Difference of squares	Διαφορά τετραγώνων	$a^2 - b^2 = (a + b)(a - b)$
Perfect square trinomial	Τέλειο τετράγωνο	$a^2 \pm 2ab + b^2 = (a \pm b)^2$
Sum / difference of cubes	Άθροισμα / διαφορά κύβων	$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$
Rational expression	Ρητή παράσταση	Ratio of two polynomials
Exponent rules	Ιδιότητες δυνάμεων	Add, subtract or multiply exponents

A ㄱ The big picture

- About 8% of Digital SAT Math questions.
- Pure algebra — no graphs, no context.
- Rewards recognizing structure over brute-force expanding.



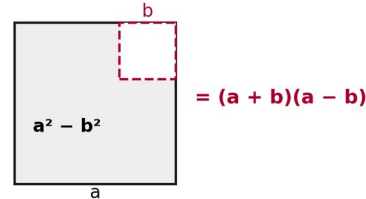
Theory Explained (1/3)

Difference of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

WHAT IT MEANS

- Works on any two squares: $4x^2 - 9 = (2x)^2 - 3^2$.
- Apply again if a factor is still a difference of squares.
- $a^2 + b^2$ does NOT factor over the reals.



Example

$$16x^4 - 1 = (4x^2 + 1)(2x + 1)(2x - 1).$$

Perfect Squares

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

WHAT IT MEANS

- The middle term $2ab$ is never missing.
- Recognize: last term a square, middle = $2 \times$ roots.
- $x^2 + 6x + 9 = (x + 3)^2$.

Example

$$(2x + 5)^2 = 4x^2 + 20x + 25.$$

Theory Explained (2/3)

Cubes

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

WHAT IT MEANS

- Binomial cube $\neq$ difference of cubes.
- Difference of cubes: PLUS ab in the second factor.
- Sum of cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$.



Example

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4).$$

Rational Expressions

factor top and bottom $\rightarrow$ cancel FACTORS

WHAT IT MEANS

- Only multiplied factors cancel — never added terms.
- Note excluded values (denominator $\neq 0$).
- $(x^2 + 4)/(x + 4)$ cannot be simplified.



Example

$$(x^2 - x - 6)/(x^2 - 3x) = (x + 2)/x.$$

Theory Explained (3/3)

Exponent Rules

$$x^a \cdot x^b = x^{a+b} \quad (x^a)^b = x^{ab} \quad x^{(m/n)} = (\sqrt[n]{x})^m$$

WHAT IT MEANS

- Multiply same bases → add exponents.
- Power of a power → multiply exponents.
- Add fractional exponents with a common denominator.



Example

$$x^{(3/2)} \cdot x^{(1/2)} = x^{(4/2)} = x^2.$$

The Substitution Check

plug in a simple value → compare every choice

WHAT IT MEANS

- Pick $x = 0, 1$ or 2 (avoid excluded values).
- Evaluate the original and each choice.
- Eliminate choices that give a different number.



Example

$$(4x^2 - 9)/(2x + 3) \text{ at } x = 0: -3 \rightarrow \text{only } 2x - 3 \text{ matches.}$$

Watch Out



$$(a + b)^2 \neq a^2 + b^2$$

- The middle term $2ab$ is always present.
- $a^2 - b^2$ factors as $(a + b)(a - b)$, not $(a - b)^2$.



Cancel factors, not terms

- $(x^2 - 4)/(x + 2) = x - 2$ only because $x^2 - 4 = (x + 2)(x - 2)$.
- Test with a number when unsure.

Problem 1

Hard

Which of the following is equivalent to $(2x + 5)^2 - (2x - 5)^2$?

- A) 0
- B) 20
- C) $40x$
- D) $4x^2 + 25$

Formula / Rule

Difference of squares: $A^2 - B^2 = (A + B)(A - B)$.

First move

Spot $A^2 - B^2$ and factor at once — faster than expanding both squares.

Solution

1. Let $A = 2x + 5$ and $B = 2x - 5$.
2. $A + B = 4x$.
3. $A - B = 10$.
4. $(A + B)(A - B) = 4x \cdot 10 = 40x$.

 **Answer:** C) $40x$

Quick check

Sum $4x \times$ difference $10 = 40x$. One line.



Problem 1 — Strategy & Elimination



Tip

Spot $A^2 - B^2$ and factor at once — faster than expanding both squares.



Shortcut

Sum $4x \times$ difference $10 = 40x$. One line.



Trap

Expanding $(2x + 5)^2$ and then mishandling the subtraction of $(2x - 5)^2$.



Desmos Tip

Graph $y = (2x+5)^2 - (2x-5)^2$ and $y = 40x$: identical lines.



Process of Elimination

Correct: **C) $40x$**

Choice	Why it is right — or exactly what went wrong
A) 0	Test $x = 1$: $49 - 9 = 40 \neq 0$.
B) 20	Test $x = 0$: $25 - 25 = 0 \neq 20$.
C) $40x$ ✓	$(4x)(10) = 40x$; $x = 1$ gives 40 ✓
D) $4x^2 + 25$	Only $(2x)^2 + 5^2$ — ignores the cross terms and the subtraction.

Problem 2

Hard

If $x \neq 0$ and $x \neq 3$, which of the following is equivalent to $(x^2 - x - 6)/(x^2 - 3x)$?

- A) $(x + 2)/x$
- B) $(x - 2)/x$
- C) $(x + 2)/(x - 3)$
- D) $1 - 2/x$

Formula / Rule

Factor numerator and denominator; cancel common FACTORS.

First move

Factor both parts fully before cancelling. Cancel factors, never terms.

Solution

1. Numerator: $x^2 - x - 6 = (x - 3)(x + 2)$.
2. Denominator: $x^2 - 3x = x(x - 3)$.
3. Cancel the common factor $(x - 3)$.
4. Result: $(x + 2)/x$.

 **Answer:** A) $(x + 2)/x$

Quick check

Test $x = 2$: original = $(-4)/(-2) = 2$; $(2 + 2)/2 = 2$ ✓



Problem 2 — Strategy & Elimination



Tip

Factor both parts fully before cancelling. Cancel factors, never terms.



Shortcut

Test $x = 2$: original = $(-4)/(-2) = 2$; $(2 + 2)/2 = 2$ ✓



Trap

Factoring only the numerator leaves $(x - 3)$ in the denominator — choice C.



Desmos Tip

Graph both expressions: identical except for holes at $x = 0$ and $x = 3$.



Process of Elimination

Correct: A) $(x + 2)/x$

Choice	Why it is right — or exactly what went wrong
A) $(x + 2)/x$ ✓	$(x - 3)(x + 2)/[x(x - 3)] = (x + 2)/x$ ✓
B) $(x - 2)/x$	Factored the numerator as $(x - 3)(x - 2)$ — wrong.
C) $(x + 2)/(x - 3)$	Didn't factor the denominator $x(x - 3)$.
D) $1 - 2/x$	Equals $(x - 2)/x$ — same error as B.

Problem 3

Hard

Which of the following is equivalent to $x^{(3/2)} \cdot x^{(1/2)}$?

- A) x^2
- B) $x^{(3/4)}$
- C) $x\sqrt{x}$
- D) x^4

Formula / Rule

Product rule: $x^a \cdot x^b = x^{a+b}$.

First move

Same base $\rightarrow$ add the exponents as fractions.

Solution

1. Add the exponents: $3/2 + 1/2$.
2. $3/2 + 1/2 = 4/2 = 2$.
3. Result: x^2 .

 **Answer:** A) x^2

Quick check

$3/2 + 1/2 = 2$, so the product is x^2 .



Problem 3 — Strategy & Elimination



Tip

Same base → add the exponents as fractions.



Trap

Multiplying the exponents instead of adding them gives $x^{(3/4)}$.



Shortcut

$3/2 + 1/2 = 2$, so the product is x^2 .



Desmos Tip

For $x > 0$, graph $y = x^{(3/2)} \cdot x^{(1/2)}$ and $y = x^2$: the same curve.



Process of Elimination

Correct: **A) x^2**

Choice	Why it is right — or exactly what went wrong
A) x^2 ✓	$3/2 + 1/2 = 2$ ✓
B) $x^{(3/4)}$	Multiplied the exponents instead of adding.
C) $x\sqrt{x}$	$= x^{(3/2)}$: only the first factor.
D) x^4	Added numerators ($3 + 1$) and dropped the denominator.

Problem 4

Hard

Which expression is equivalent to $(3x - 2)^3$?

- A) $27x^3 - 8$
- B) $27x^3 - 54x^2 + 36x - 8$
- C) $27x^3 + 54x^2 - 36x - 8$
- D) $9x^3 - 6x^2 + 4x - 8$

Formula / Rule

Binomial cube: $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

First move

Signs in $(a - b)^3$ alternate: +, -, +, -.

Solution

1. Let $a = 3x$ and $b = 2$.
2. $a^3 = 27x^3$.
3. $3a^2b = 3(9x^2)(2) = 54x^2$.
4. $3ab^2 = 3(3x)(4) = 36x$.
5. $b^3 = 8$.
6. Result: $27x^3 - 54x^2 + 36x - 8$.

 **Answer:** B) $27x^3 - 54x^2 + 36x - 8$

Quick check

$x = 1$: $(3 - 2)^3 = 1$; only B gives $27 - 54 + 36 - 8 = 1$.



Problem 4 — Strategy & Elimination



Tip

Signs in $(a - b)^3$ alternate: +, −, +, −.



Trap

$(a - b)^3 \neq a^3 - b^3$ — the two middle terms cannot be dropped.



Shortcut

$x = 1$: $(3 - 2)^3 = 1$; only B gives $27 - 54 + 36 - 8 = 1$.



Desmos Tip

Graph $y = (3x-2)^3$ and choice B: identical curves.



Process of Elimination

Correct: **B) $27x^3 - 54x^2 + 36x - 8$**

Choice	Why it is right — or exactly what went wrong
A) $27x^3 - 8$	That is $a^3 - b^3$, not $(a - b)^3$.
B) $27x^3 - 54x^2 + 36x - 8$ ✓	Correct binomial expansion ✓
C) $27x^3 + 54x^2 - 36x - 8$	Sign errors on the middle terms.
D) $9x^3 - 6x^2 + 4x - 8$	Used $(3x)^3 = 9x^3$ instead of $27x^3$.

Problem 5

Hard

Which of the following is equivalent to $(4x^2 - 9)/(2x + 3)$?

- A) $2x + 3$
- B) $2x - 3$
- C) $4x - 9$
- D) $2x$

Formula / Rule

Difference of squares: $4x^2 - 9 = (2x + 3)(2x - 3)$.

First move

Spot the difference of squares in the numerator immediately.

Solution

1. Recognize $4x^2 - 9 = (2x)^2 - 3^2$.
2. Factor: $(2x + 3)(2x - 3)$.
3. Cancel $(2x + 3)$.
4. Result: $2x - 3$.

 **Answer:** B) $2x - 3$

Quick check

$x = 0$: $-9/3 = -3$; only $2x - 3$ gives -3 .



Problem 5 — Strategy & Elimination



Tip

Spot the difference of squares in the numerator immediately.



Trap

Writing the cancelled factor $(2x + 3)$ as the answer.



Shortcut

$x = 0$: $-9/3 = -3$; only $2x - 3$ gives -3 .



Desmos Tip

Graph both: identical, with a hole at $x = -3/2$.



Process of Elimination

Correct: **B) $2x - 3$**

Choice	Why it is right — or exactly what went wrong
A) $2x + 3$	That is the factor that cancels.
B) $2x - 3$ ✓	$(2x + 3)(2x - 3)/(2x + 3) = 2x - 3$ ✓
C) $4x - 9$	Divided term by term — invalid.
D) $2x$	Ignored the constants.

Problem 6

Hard

The expression $16x^4 - 1$ can be written as the product of three factors. What are those factors?

- A) $(4x^2 + 1)(2x + 1)(2x - 1)$
- B) $(4x^2 + 1)(4x^2 - 1)$
- C) $(2x - 1)^2(2x + 1)^2$
- D) $(4x + 1)(4x - 1)(2x + 1)$

Formula / Rule

Apply the difference of squares twice.

First move

After factoring once, check whether a factor is ALSO a difference of squares.

Solution

1. Write $16x^4 - 1 = (4x^2)^2 - 1^2$.
2. Factor: $(4x^2 + 1)(4x^2 - 1)$.
3. $4x^2 - 1$ is again a difference of squares: $(2x + 1)(2x - 1)$.
4. Result: $(4x^2 + 1)(2x + 1)(2x - 1)$.

 **Answer:** A) $(4x^2 + 1)(2x + 1)(2x - 1)$

Quick check

$$16x^4 - 1 \rightarrow (4x^2 + 1)(4x^2 - 1) \rightarrow (4x^2 + 1)(2x + 1)(2x - 1).$$



Problem 6 — Strategy & Elimination



Tip

After factoring once, check whether a factor is ALSO a difference of squares.



Shortcut

$16x^4 - 1 \rightarrow (4x^2 + 1)(4x^2 - 1) \rightarrow (4x^2 + 1)(2x + 1)(2x - 1).$



Trap

Stopping after one step (choice B) — only two factors.



Desmos Tip

Graph $y = 16x^4 - 1$ and choice A: identical.



Process of Elimination

Correct: A) $(4x^2 + 1)(2x + 1)(2x - 1)$

Choice	Why it is right — or exactly what went wrong
A) $(4x^2 + 1)(2x + 1)(2x - 1)$ ✓	Three factors, multiplies back to $16x^4 - 1$ ✓
B) $(4x^2 + 1)(4x^2 - 1)$	Only two factors — $4x^2 - 1$ factors further.
C) $(2x - 1)^2(2x + 1)^2$	$= (4x^2 - 1)^2 = 16x^4 - 8x^2 + 1.$
D) $(4x + 1)(4x - 1)(2x + 1)$	Degree 3 — cannot equal a degree-4 expression.

Problem 7

Very Hard

Which of the following is equivalent to $(x^2 + 4x + 4)/(x^2 + 5x + 6)$?

- A) $(x + 2)/(x + 3)$
- B) $(x + 2)/(x - 3)$
- C) $(x - 2)/(x + 3)$
- D) $4/6$

Formula / Rule

Factor both parts; cancel common factors.

First move

Recognize the perfect-square numerator: $(x + 2)^2$.

Solution

1. Numerator: $x^2 + 4x + 4 = (x + 2)^2$.
2. Denominator: $x^2 + 5x + 6 = (x + 2)(x + 3)$.
3. Cancel one factor of $(x + 2)$.
4. Result: $(x + 2)/(x + 3)$.

 **Answer:** A) $(x + 2)/(x + 3)$

Quick check

$x = 0$: $4/6 = 2/3$; $(0 + 2)/(0 + 3) = 2/3$ ✓



Problem 7 — Strategy & Elimination



Tip

Recognize the perfect-square numerator: $(x + 2)^2$.



Trap

'Simplifying' by comparing constants ($4/6$) — invalid for polynomials.



Shortcut

$x = 0$: $4/6 = 2/3$; $(0 + 2)/(0 + 3) = 2/3$ ✓



Desmos Tip

Graph both: identical, with a hole at $x = -2$.



Process of Elimination

Correct: **A)** $(x + 2)/(x + 3)$

Choice	Why it is right — or exactly what went wrong
A) $(x + 2)/(x + 3)$ ✓	$(x + 2)^2/[(x + 2)(x + 3)]$ ✓
B) $(x + 2)/(x - 3)$	Wrong sign: $x^2 + 5x + 6 = (x + 2)(x + 3)$.
C) $(x - 2)/(x + 3)$	Wrong factoring of the numerator.
D) $4/6$	Compared constant terms — invalid.

Problem 8

Very Hard

The expression $(x^3 - 8)/(x - 2)$ is equivalent to which of the following?

- A) $x^2 + 4$
- B) $x^2 - 4$
- C) $x^2 + 2x + 4$
- D) $x^2 - 2x + 4$

Formula / Rule

Difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

First move

In $a^3 - b^3$, the second factor has a PLUS ab term.

Solution

1. Write $x^3 - 8 = x^3 - 2^3$.
2. Factor: $(x - 2)(x^2 + 2x + 4)$.
3. Cancel $(x - 2)$.
4. Result: $x^2 + 2x + 4$.

 **Answer:** C) $x^2 + 2x + 4$

Quick check

Test $x = 1$: $(1 - 8)/(1 - 2) = 7$. Only C gives $1 + 2 + 4 = 7$.



Problem 8 — Strategy & Elimination



Tip

In $a^3 - b^3$, the second factor has a PLUS ab term.



Shortcut

Test $x = 1$: $(1 - 8)/(1 - 2) = 7$. Only C gives $1 + 2 + 4 = 7$.



Trap

Choice D uses $-2x$ — that is the sum-of-cubes pattern.



Desmos Tip

Graph both: identical, with a hole at $x = 2$.



Process of Elimination

Correct: C) $x^2 + 2x + 4$

Choice	Why it is right — or exactly what went wrong
A) $x^2 + 4$	Dropped the middle term.
B) $x^2 - 4$	$(x - 2)(x^2 - 4)$ is not $x^3 - 8$.
C) $x^2 + 2x + 4$ ✓	$(x - 2)(x^2 + 2x + 4) = x^3 - 8$ ✓
D) $x^2 - 2x + 4$	Sum-of-cubes pattern (for $x^3 + 8$).

Advanced Math Cheat Sheet

01 Nonlinear Functions

- ▶ Vertex $x = -b/(2a)$
- ▶ $a(x - h)^2 + k \rightarrow$ vertex (h, k)
- ▶ Zeros from factored form
- ▶ $a \cdot b^x$: $a = f(0)$, $r = b - 1$
- ▶ $f(x - h) \rightarrow$ right h · $f(x) + k \rightarrow$ up k
- ▶ $-f(x)$: reflect over x-axis

02 Nonlinear Equations & Systems

- ▶ $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$
- ▶ $\Delta > 0$: 2 · $\Delta = 0$: 1 · $\Delta < 0$: 0
- ▶ Sum = $-b/a$ · Product = c/a
- ▶ Line & parabola: substitute, check Δ
- ▶ Radicals: square, then CHECK

03 Equivalent Expressions

- ▶ $a^2 - b^2 = (a + b)(a - b)$
- ▶ $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- ▶ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- ▶ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- ▶ Cancel factors only
- ▶ Check with $x = 0, 1, 2$

★ Golden Rules

- ▶ Vertex form $a(x - h)^2 + k$ reveals the vertex, axis and direction instantly.
- ▶ $\Delta = b^2 - 4ac$ counts the real solutions without solving.
- ▶ Vieta gives the sum ($-b/a$) or product (c/a) in one step.
- ▶ In rational expressions, FACTOR first, then cancel factors — never terms.
- ▶ Check equivalent-expression choices by substituting a simple value.
- ▶ In $a \cdot b^x$, the rate is $b - 1$ — never confuse rate and factor.
- ▶ After squaring or clearing denominators, verify in the ORIGINAL equation.

Top Traps to Avoid

Skill	The trap	The fix
Nonlinear Functions	Vertex x is $-b/(2a)$, not $-b/a$	The missing $\div 2$ is the #1 arithmetic error. In $a(x - h)^2$, h is subtracted: $(x - 3)^2 \rightarrow x = +3$.
Nonlinear Equations & Systems	Check in the ORIGINAL equation	After squaring or clearing denominators, some roots are fake. Extraneous roots appear as answer choices on purpose.
Equivalent Expressions	$(a + b)^2 \neq a^2 + b^2$	The middle term $2ab$ is always present. $a^2 - b^2$ factors as $(a + b)(a - b)$, not $(a - b)^2$.



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